

DATA CLEANING & PREPARATION

Audit Context

This document is the official accountability record for all transformations applied to the raw e-commerce order dataset as part of DecodeLabs Project 1. Every change is logged with its Change ID, the phase it belongs to, a precise description of the action taken, the measurable impact on the dataset, and its resolution status. No data was fabricated — all figures are sourced directly from the cleaning pipeline output.

1200 Raw Rows Ingested	1200 Clean Rows Output	309 Missing Values (Before)	0 Missing Values (After)	9 Total Changes Logged
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Change Log

ID	Phase	Description	Impact	Status
CR000	Ingestion	Loaded raw dataset	1200 rows × 14 columns ingested	Resolved
CR001	Phase 2 Integrity Audit	Audited for fully duplicated rows	0 exact duplicates found	Resolved
CR002	Phase 2 Integrity Audit	Audited OrderID uniqueness — Verification Gate PASSED	0 duplicate OrderIDs (0% error rate)	Resolved
CR003	Phase 1 Strategic Imputation	Imputed missing CouponCode as 'NoCoupon' — categorical placeholder, NOT deleted	Preserved 309 records (listwise deletion would have lost them)	Resolved
CR004	Phase 3 Format Sta ndardisatio n	Standardised Date to ISO 8601 (YYYY-MM-DD) — Verification Gate PASSED	0 invalid dates (0% error rate)	Resolved
CR005	Phase 3 Format Sta ndardisatio n	Trimmed whitespace & applied Title Case to text/categorical fields	Standardised 5 columns	Resolved
CR006	Phase 3 Format Sta ndardisatio n	Trimmed whitespace & standardised UPPER CASE on identifier/code fields	Standardised 4 columns	Resolved
CR007	Phase 3 Format Sta ndardisatio n	Enforced 2-decimal numeric precision on price fields	Standardised UnitPrice, TotalPrice	Resolved
CR008	Phase 4 Cross-field Validation	Validated TotalPrice = Quantity × UnitPrice for every record	0 calculation mismatches found	Resolved

Verification Gate Results

Metric	Target	Result	Gate
Duplicate Order IDs	0	0	PASSED ✓
Invalid / Malformed Dates	0	0	PASSED ✓
Missing Values Remaining	0	0	PASSED ✓
Rows Lost During Cleaning	0	0	PASSED ✓